전산화학 (2010년 1학기)

1차 중간고사 정답

(**총 11 문제, 140점**)

1. **(총10점, 각5점)**

(a)

$$\begin{aligned} z &= (x+iy)^2 - 4e^{ixy} \\ &= x^2 + 2xyi - y^2 - 4(\cos(xy) + i\sin(xy) \\ &= (x^2 - y^2 - 4\cos(xy)) + i(2xy - 4\sin(xy)) \end{aligned}$$

그러므로 $z^* = (x^2 - y^2 - \cos(xy)) - i(2xy - 4\sin(xy))$ 이고, $\mathrm{Re}(z^*) = x^2 - y^2 - 4\cos(xy)$ 이고 $\mathrm{Im}(z^*) = -2xy + 4\sin(xy)$ 이다.

(b)

$$\begin{aligned} z &= (\sqrt{3} - 3i)e^{-5\pi/3} \\ &= (\sqrt{3} - 3i)\left(\cos\left(-\frac{5\pi}{3}\right) + i\sin\left(-\frac{5\pi}{3}\right)\right) \\ &= (\sqrt{3} - 3i)\left(\cos\left(\frac{5\pi}{3}\right) - i\sin\left(\frac{5\pi}{3}\right)\right) \\ &= (\sqrt{3} - 3i)\left(\cos\left(2\pi - \frac{\pi}{3}\right) - i\sin\left(2\pi - \frac{\pi}{3}\right)\right) \\ &= (\sqrt{3} - 3i)\left(\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right) \\ &= (\sqrt{3} - 3i)\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) \\ &= \frac{\sqrt{3}}{2} + \frac{3}{2}i - \frac{3}{2}i + \frac{3\sqrt{3}}{2} \\ &= \frac{\sqrt{3}}{2} + \frac{3\sqrt{3}}{2} = 2\sqrt{3} \end{aligned}$$

그러므로 $z^* = 2\sqrt{3}$ 이고, $\mathrm{Re}(z^*) = 2\sqrt{3}$ 이고 $\mathrm{Im}(z^*) = 0$ 이다.

2. **(총10점, 각5점)**

(a)

$$
\begin{aligned}
f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y &= 2axe^{ax^2+by^2} \\
f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x &= 2bye^{ax^2+by^2} \\
f_{xx}(x,y) = \left(\frac{\partial^2 f}{\partial x^2}\right)_y &= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)_y = 2ae^{ax^2+by^2} + 4a^2x^2e^{ax^2+by^2} \\
f_{yy}(x,y) = \left(\frac{\partial^2 f}{\partial y^2}\right)_x &= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)_x = 2be^{ax^2+by^2} + 4b^2y^2e^{ax^2+by^2} \\
f_{xy}(x,y) = \left(\frac{\partial^2 f}{\partial y \partial x}\right) &= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)_y = 4abxye^{ax^2+by^2} \\
f_{yx}(x,y) = \left(\frac{\partial^2 f}{\partial x \partial y}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)_x = 4abxye^{ax^2+by^2}
\end{aligned}
$$

(b)

$$
\begin{aligned}
f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y = e^{-y}\big(-y\sin(xy)\big) &= -ye^{-y}\sin(xy) \\
f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x = -e^{-y}\big(-x\sin(xy)\big) &= -e^{-y}\cos(xy) - xe^{-y}\sin(xy) \\
f_{xx}(x,y) = \left(\frac{\partial^2 f}{\partial x^2}\right)_y = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)_y &= -y^2e^{-y}\cos(xy) \\
f_{yy}(x,y) = \left(\frac{\partial^2 f}{\partial y^2}\right)_x = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)_x &= e^{-y}\cos(xy) + xe^{-y}\sin(xy) + xe^{-y}\sin(xy) - x^2e^{-y}\cos(xy \\
&= e^{-y}\cos(xy) + 2xe^{-y}\sin(xy) - x^2e^{-y}\cos(xy) \\
f_{xy}(x,y) = \left(\frac{\partial^2 f}{\partial y \partial x}\right) = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)_y &= -e^{-y}\sin(xy) + ye^{-y}\sin(xy) - xye^{-y}\cos(xy) \\
&= e^{-y}\big(-\sin(xy) + y\sin(xy) - xy\cos(xy)\big) \\
f_{yx}(x,y) = \left(\frac{\partial^2 f}{\partial x \partial y}\right) = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)_x &= ye^{-y}\sin(xy) - e^{-y}\sin(xy) - xye^{-y}\cos(xy) \\
&= e^{-y}\big(-\sin(xy) + y\sin(xy) - xy\cos(xy)\big)
\end{aligned}
$$

3. **(총 10점, 각 5점)**

(a)

$$\left(\frac{\partial M}{\partial V}\right)_T = -\frac{RT}{(V-b)^2}$$

$$\left(\frac{\partial N}{\partial T}\right)_V = -\frac{RT}{(V-b)^2} - \frac{2a}{T^2V^2}$$

$(\partial M/\partial V)_T \neq (\partial N/\partial T)_V$ 0이므로 **완전미분이 아니다.**

(b)

$$\left(\frac{\partial M}{\partial V}\right)_T = 0$$

$$\left(\frac{\partial N}{\partial T}\right)_V = \frac{R}{V}$$

$(\partial M/\partial V)_T \neq (\partial N/\partial T)_V$ 이므로 **완전미분이 아니다.**

4. **(총 10점, 각 5점)**

(a) $f(x,y)$가 (x^*, y^*)에서 극값을 가지므로

$$\begin{aligned} f_x(x^*, y^*) &= 0 \Rightarrow \frac{2x^*}{x^{*2} + y^{*2} + 2} = 0 \Rightarrow x^* = 0 \\ f_y(x^*, y^*) &= 0 \Rightarrow \frac{2y^*}{x^{*2} + y^{*2} + 2} = 0 \Rightarrow y^* = 0 \end{aligned}$$

그러므로 $x^* = 0,\ y^* = 0$

(b) Hessian 검사를 이용하면

$$\Delta_1 = f_{xx}(x^*, y^*) \quad \text{그리고} \quad \Delta_2 = f_{xx}(x^*, y^*) f_{yy}(x^*, y^*) - f_{xy}(x^*, y^*)^2.$$

주어진 함수 $f(x,y) = 4 + x + y - x^2 - xy - \frac{y^2}{2}$에 대하여,

$$\begin{aligned} f_{xx}(x,y) &= \frac{2(x^2 + y^2 + 2) - 2x(2x)}{(x^2 + y^2 + 2)^2} = \frac{-2x^2 + 2y^2 + 4}{(x^2 + y^2 + 2)^2} \\ f_{yy}(x,y) &= \frac{2(x^2 + y^2 + 2) - 2y(2y)}{(x^2 + y^2 + 2)^2} = \frac{2x^2 - 2y^2 + 4}{(x^2 + y^2 + 2)^2} \\ f_{xy}(x,y) &= \frac{-2x(2y)}{(x^2 + y^2 + 2)^2} = \frac{-4xy}{(x^2 + y^2 + 2)^2} \end{aligned}$$

그러므로

$$\begin{aligned} \Delta_1 &= f_{xx}(x^*, y^*) = f_{xx}(0,0) = 1 > 0 \\ \Delta_2 &= f_{xx}(x^*, y^*) f_{yy}(x^*, y^*) - f_{xy}(x^*, y^*)^2 = f_{xx}(0,0) f_{yy}(0,0) - f_{xy}(0,0)^2 = 1 \times 1 - 0 = 1 > 0 \end{aligned}$$

$\Delta_1 > 0$ 이고 $\Delta_2 > 0$ 이므로 $f(x,y)$는 $x^* = 0,\ y^* = 0$에서 극소값을 갖는다.

5. **(총 15점)**

(a) **(5점)**

$$I = \int_{-\infty}^{\infty} dx e^{-a^2 x^2} = \int_{-\infty}^{\infty} dy e^{-a^2 y^2}. \tag{1}$$

$$
\begin{aligned}
I^2 &= \int_{-\infty}^{\infty} dx e^{-a^2x^2} \times \int_{-\infty}^{\infty} dy e^{-a^2y^2}, \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} dxdy e^{-a^2(x^2+y^2)}, \\
&\quad \text{직교 좌표의 극 좌표 변환을 이용하면} \\
&= \int_0^{2\pi}\int_0^{\infty} rdrd\theta e^{-a^2r^2}, \\
&\quad t = ar \text{ 로 치환하면,} \\
&= \frac{1}{a^2}\int_0^{2\pi} d\theta \int_0^{\infty} dtte^{-t^2}, \\
&= \frac{1}{a^2}2\pi\Big[-\frac{1}{2}e^{-t^2}\Big]_0^{\infty} \\
&= \frac{\pi}{a^2}
\end{aligned}
$$

$$\therefore I = \frac{\sqrt{\pi}}{a}. \tag{2}$$

(b) **(10점)** $a = (1-i)/\sqrt{2}$ 이므로

$$a^2 = \frac{1-i}{\sqrt{2}}\frac{1-i}{\sqrt{2}} = \frac{1-2i-1}{2} = -i \tag{3}$$

(3) 식을 $\int_{-\infty}^{\infty} e^{-a^2x^2}dx$ 에 넣으면,

$$\int_{-\infty}^{\infty} e^{-a^2x^2}dx = \int_{-\infty}^{\infty} e^{ix^2}dx = \int_{-\infty}^{\infty} \cos x^2 dx + i\int_{-\infty}^{\infty} \sin x^2 dx \tag{4}$$

한편, (a)의 (2) 식으로부터 $f(a) = \sqrt{\pi}/a$ 이므로

$$\frac{\sqrt{\pi}}{a} = \frac{\sqrt{\pi}}{(1-i)/\sqrt{2}} = \frac{\sqrt{2\pi}}{1-i} = \frac{\sqrt{2\pi}}{1-i}\times\frac{1+i}{1+i} = \frac{\sqrt{2\pi}}{2}(1+i) \tag{5}$$

(4) 식과 (5) 식을 비교하고 복소수 상등 정리를 이용하면

$$
\begin{aligned}
\int_{-\infty}^{\infty} \cos x^2 dx &= \frac{\sqrt{2\pi}}{2} \\
\int_{-\infty}^{\infty} \sin x^2 dx &= \frac{\sqrt{2\pi}}{2}
\end{aligned}
$$

6. **(총 10점, 각 5점)**

(a) $\int x^2 \ln x dx$

만약 $u = \ln x$ 그리고 $v' = x^2dx$ 라면, $u' = \frac{dx}{x}$ 그리고 $v = \frac{1}{3}x^3$ 이므로,

$$
\begin{aligned}
\int x^2 \ln x dx &= \frac{1}{3}x^3\ln x - \int \frac{1}{3}x^3\frac{1}{x}dx \\
&= \frac{1}{3}x^3\ln x - \frac{1}{3}\int x^2 dx \\
&= \frac{1}{3}x^3\ln x - \frac{1}{9}x^3 + C.
\end{aligned}
$$

(b)

$$\begin{aligned}\int x\sin^2\left(\frac{N\pi x}{A}\right)dx &= \int\frac{1}{2}\left(1-\cos\left(\frac{2N\pi x}{A}\right)\right)xdx,\\ &= \int\left(\frac{1}{2}x-\frac{1}{2}x\cos\left(\frac{2N\pi x}{A}\right)\right)dx,\\ &= \frac{1}{4}x^2-\frac{1}{2}\int x\cos\left(\frac{2N\pi x}{A}\right)dx. \qquad (6)\end{aligned}$$

If $u=x$ and $v'=\cos\left(\frac{2N\pi x}{A}\right)dx$, then $u'=dx$ and $u=\frac{A}{2N\pi}\sin\left(\frac{2N\pi x}{A}\right)$ Eq. (6) is given by

$$\begin{aligned}\frac{1}{4}x^2-\int x\cos\left(\frac{2N\pi x}{A}\right)dx &= \frac{1}{4}x^2-\frac{1}{2}\left\{x\frac{A}{2N\pi}\sin\left(\frac{2N\pi x}{A}\right)-\int\frac{A}{2N\pi}\sin\left(\frac{2N\pi x}{A}\right)\right\},\\ &= \frac{1}{4}x^2-\frac{A}{4N\pi}x\sin\left(\frac{2N\pi x}{A}\right)+\frac{A}{4N\pi}\int\sin\left(\frac{2N\pi x}{A}\right),\\ &= \frac{1}{4}x^2-\frac{A}{4N\pi}x\sin\left(\frac{2N\pi x}{A}\right)-\frac{A^2}{8N^2\pi^2}\cos\left(\frac{2N\pi x}{A}\right)+C.\end{aligned}$$

7. **(총 10점, 각 5점)**

(a) If $\frac{n\pi x}{a}=t$, $dx=\frac{a}{n\pi}dt$ and

$$\int_0^a x^2\sin\left(\frac{n\pi x}{a}\right)dx = \left(\frac{a}{n\pi}\right)^3\int_0^{n\pi}t^2\sin t\,dt \qquad (7)$$

부분 적분법을 이용하면 $u=t^2$ 그리고 $v'=\sin(t)dt$, 그러면 $u'=2tdt$, $v=-\cos(t)$, 그러므로 (7) 식은 다음과 같이 쓸 수 있다.

$$\begin{aligned}\left(\frac{a}{n\pi}\right)^3\int_0^{n\pi}t^2\sin(t)dt &= \left(\frac{a}{n\pi}\right)^3\left\{\Big[-t^2\cos(t)\Big]_0^{n\pi}+2\int_0^{n\pi}t\cos(t)dt\right\},\\ &= \left(\frac{a}{n\pi}\right)\left\{-(n\pi)^2\cos(n\pi)+2\int_0^{n\pi}t\cos(t)dt\right\}, \qquad (8)\end{aligned}$$

(8) 식의 두 번째 항에 대하여 다시 부분 적분법을 이용하면, $u=t$ 그리고 $v'=\cos(t)dt$, 그러면 $u'=dt$, $v=\sin(t)$ 이므로 (8) 식은 다음과 같이 쓸 수 있다.

$$\begin{aligned}\left(\frac{a}{n\pi}\right)^3\int_0^{n\pi}t^2\sin(t)dt &= -\frac{a^3}{n\pi}\cos(n\pi)+2\left(\frac{a}{n\pi}\right)^3\left\{\Big[t\sin(t)\Big]_0^{n\pi}-\int_0^{n\pi}\sin(t)dt\right\},\\ &\qquad \sin(n\pi)=0 \text{ 이므로}\\ &= -\frac{a^3}{n\pi}\cos(n\pi)+2\left(\frac{a}{n\pi}\right)^3\Big[\cos(t)\Big]_0^{n\pi}\\ &= -\frac{a^3}{n\pi}\cos(n\pi)+2\left(\frac{a}{n\pi}\right)^3\Big(\cos(n\pi)-1\Big)\\ &= -\frac{a^3}{n\pi}(-1)^n+2\left(\frac{a}{n\pi}\right)^3\Big((-1)^n-1\Big) \qquad (\cos(n\pi)=(-1)^n)\end{aligned}$$

(b) $\int_0^{\pi}(3\cos^3\theta-1)^2\sin\theta d\theta$ 에서 $t=\cos\theta$ 라 치환하면 $dt=-\sin\theta d\theta$ 이고 식은 다음과 같이 쓸 수 있다.

$$\begin{aligned}\int_0^{\pi}(3\cos^3\theta-1)^2\sin\theta d\theta &= \int_1^{-1}(3t^3-1)^2(-dt)\\ &= \int_{-1}^{1}(3t^3-1)^2dt\\ &= \int_{-1}^{1}(9t^6-6t^3+1)^2dt\\ &= \left[\frac{9}{7}t^7-\frac{6}{4}t^4+t\right]_{-1}^{1}\\ &= \frac{9}{7}-\frac{6}{4}+1-\left(-\frac{9}{7}-\frac{6}{4}-1\right)\\ &= \frac{18}{7}+2=\frac{32}{7}\end{aligned}$$

8. **(총 10점, 각 5점)**

(a)

$$\int_0^{\infty}e^{-ax}dx=\left[-\frac{1}{a}e^{-ax}\right]_0^{\infty}=\frac{1}{a} \tag{9}$$

(9) 식의 양 변을 a 에 대해 미분하여 정리하면

$$\int_0^{\infty}xe^{-ax}dx=\frac{1}{a^2} \tag{10}$$

(10) 식의 양 변을 a 에 대해 미분하여 정리하면

$$\int_0^{\infty}x^2e^{-ax}dx=\frac{2}{a^3} \tag{11}$$

(11) 식의 양 변을 a 에 대해 미분하여 정리하면

$$\int_0^{\infty}x^3e^{-ax}dx=\frac{6}{a^4} \tag{12}$$

(b)

$$\begin{aligned}&\left(\frac{1}{a_0}\right)^3\int_0^{\infty}r\left(2-\frac{r}{a_0}\right)^2e^{-r/a_0}dr\\ =\ &\left(\frac{1}{a_0}\right)^3\left(\int_0^{\infty}4re^{-r/a_0}dr-\int_0^{\infty}\frac{4r^2}{a_0}e^{-r/a_0}+\int_0^{\infty}\frac{r^3}{a_0^2}e^{-r/a_0}dr\right)\\ =\ &\frac{4}{a_0^3}\int_0^{\infty}drre^{-r/a_0}-\frac{4}{a_0^4}\int_0^{\infty}drr^2e^{-r/a_0}+\frac{1}{a_0^5}\int_0^{\infty}drr^3e^{-r/a_0}\\ &\text{(10) 식, (11) 식, 그리고 (12) 식을 이용하면}\\ =\ &\frac{4}{a_0^3}(a_0^2)-\frac{4}{a_0^4}(2a_0^3)+\frac{1}{a_0^5}(6a_0^4)\\ =\ &\frac{2}{a_0}\end{aligned}$$

9. **(총 20점, 각 5점)**

(a)

$$\int_C (y\cos x dx + \sin x dy),$$

위 식에서 경로 C는 점 $(0,0)$부터 점 $(\pi/2, \pi/2)$을 연결하는 직선 $y - x = 0$을 의미한다.

$$\begin{aligned}
\int_C (y\cos x dx + \sin x dy) &= \int_0^{\pi/2} x\cos x dx + \int_0^{\pi/2} \sin y dy \\
&= \Big[x\sin x\Big]_0^{\pi/2} - \int_0^{\pi/2} \sin x dx + \Big[-\cos y\Big]_0^{\pi/2} \\
&= \frac{\pi}{2} - \Big[-\cos x\Big]_0^{\pi/2} + 1 \\
&= \frac{\pi}{2}
\end{aligned}$$

(b)

$$\int_C (y\cos x dx + \sin x dy),$$

위 식에서 경로 C는 처음에는 $(0,0)$에서 $(\pi/2, 0)$까지 그 다음에는 $(\pi/2, 0)$에서 $(\pi/2, \pi/2)$로 주어진다.

$$\begin{aligned}
\int_C (y\cos x dx + \sin x dy) &= \int_0^{\pi/2} 0 \times \cos x dx + \sin\left(\frac{\pi}{2}\right) \int_0^{\pi/2} dy \\
&= \frac{\pi}{2}.
\end{aligned}$$

(c) $M(x,y) = y\cos x,\ N(x,y) = \sin x$ 이므로

$$\left(\frac{\partial M}{\partial y}\right)_x = \cos x, \qquad \left(\frac{\partial N}{\partial x}\right)_y = \cos x \Rightarrow \left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y$$

이므로, $y\cos x dx + \sin x dy$는 완전미분이다.

(d) $C : (0,0) \to (\pi/2, \pi/2)$

$$\int_C (y\cos x dx + \sin x dy) = \Big[y\sin x\Big]_{(0,0)}^{(\pi/2,\pi/2)} = \frac{\pi}{2}\sin\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$$

10. **(총 25점, 각 5점)**

(a)

$$dG(T,P) = \left(\frac{\partial G}{\partial T}\right)_P dT + \left(\frac{\partial G}{\partial P}\right)_T dP$$

위 식을 문제 10 번의 (1) 식과 비교하면

$$\left(\frac{\partial G}{\partial T}\right)_P = -S \tag{13}$$

그리고

$$\left(\frac{\partial G}{\partial P}\right)_T = V \tag{14}$$

(b) $G = H - TS$ 이므로 $S = \frac{H-G}{T}$. 이 식을 (a)의 (13) 식에 대입하면

$$\left(\frac{\partial G}{\partial T}\right)_P = \frac{G-H}{T} \tag{15}$$

(c)

$$\begin{aligned}\left(\frac{\partial}{\partial T}\frac{G}{T}\right)_P &= -\frac{G}{T^2} + \frac{1}{T}\left(\frac{\partial G}{\partial T}\right)_P \\ &= -\frac{G}{T^2} + \frac{1}{T}\frac{G-H}{T} \\ &= -\frac{H}{T^2}\end{aligned} \tag{16}$$

(d)

$$\left[\frac{\partial(\Delta H)}{\partial T}\right]_P = \Delta C_P = a + bT + cT^2. \tag{17}$$

ΔH 를 구하기위해 식 (17)를 온도 T에 대해 적분하면

$$\Delta H = aT + \frac{b}{2}T^2 + \frac{c}{3}T^3 + d \tag{18}$$

윗식에서 d는 적분상수이다.

(e) (c)의 (16) 식에서 G를 ΔG 그리고 H를 ΔH로 치환하면,

$$\left(\frac{\partial}{\partial T}\frac{\Delta G}{T}\right)_P = -\frac{\Delta H}{T^2} \tag{19}$$

$\Delta H = aT + \frac{b}{2}T^2 + \frac{c}{3}T^3 + d$ 이므로,

$$\left[\frac{\partial(\Delta G/T)}{\partial T}\right]_p = -\frac{\Delta H}{T^2} = -\frac{a}{T} - \frac{b}{2} - \frac{c}{3}T - \frac{d}{T^2}.$$

그러므로,

$$\begin{aligned}\int\left[\frac{\partial(\Delta G/T)}{\partial T}\right]_p dT = \frac{\Delta G}{T} &= \int\left(-\frac{a}{T} - \frac{b}{2} - \frac{c}{3}T - \frac{d}{T^2}\right)dT, \\ &= -a\ln T - \frac{b}{2}T - \frac{c}{6}T^2 + \frac{d}{T} + e,\end{aligned}$$

윗식에서 e는 적분상수이다. 그러므로,

$$\Delta G = -aT\ln T - \frac{b}{2}T^2 - \frac{c}{6}T^3 + d + eT.$$

11. **(총 10점, 각 5점)**

(a)

$$
\begin{aligned}
& \int_0^{\pi} d\theta \int_0^{2\pi} d\phi Y_1^0(\theta,\phi) Y_2^0(\theta,\phi) \sin\theta \\
= & \int_0^{\pi} d\theta \int_0^{2\pi} d\phi \left(\frac{3}{4\pi}\right)^{1/2} \cos\theta \left(\frac{5}{16\pi}\right)^{1/2} (3\cos^2\theta - 1)\sin\theta \\
= & \left(\frac{3}{4\pi}\right)^{1/2} \left(\frac{5}{16\pi}\right)^{1/2} \int_0^{2\pi} d\phi \int_0^{\pi} \cos\theta (3\cos^2\theta - 1)\sin\theta d\theta \\
= & \left(\frac{3}{4\pi}\right)^{1/2} \left(\frac{5}{16\pi}\right)^{1/2} (2\pi) \int_0^{\pi} \cos\theta (3\cos^2\theta - 1)\sin\theta d\theta
\end{aligned}
$$

$t = \cos\theta$ 라고 치환하면, $dt = -\sin\theta d\theta$

$$
\begin{aligned}
= & \left(\frac{3}{4\pi}\right)^{1/2} \left(\frac{5}{16\pi}\right)^{1/2} (2\pi) \int_1^{-1} t(3t^2 - 1) \\
= & \left(\frac{3}{4\pi}\right)^{1/2} \left(\frac{5}{16\pi}\right)^{1/2} (2\pi) \int_{-1}^{1} (3t^3 - t)(dt)
\end{aligned}
$$

$3t^3 - t$ 는 기함수0이므로

$$= 0$$

(b)

$$
\begin{aligned}
& \int_0^{\pi} d\theta \int_0^{2\pi} d\phi Y_1^0(\theta,\phi)(\cos\theta)Y_2^0(\theta,\phi)\sin\theta \\
= & \int_0^{\pi} d\theta \int_0^{2\pi} d\phi \left(\frac{3}{4\pi}\right)^{1/2} \cos\theta(\cos\theta)\left(\frac{5}{16\pi}\right)^{1/2}(3\cos^2\theta-1)\sin\theta \\
= & \left(\frac{3}{4\pi}\right)^{1/2}\left(\frac{5}{16\pi}\right)^{1/2}\int_0^{2\pi} d\phi \int_0^{\pi} \cos^2\theta(3\cos^2\theta-1)\sin\theta d\theta \\
= & \left(\frac{3}{4\pi}\right)^{1/2}\left(\frac{5}{16\pi}\right)^{1/2}(2\pi)\int_0^{\pi} \cos^2\theta(3\cos^2\theta-1)\sin\theta d\theta
\end{aligned}
$$

$t=\cos\theta$ 라고 치환하면, $dt=-\sin\theta d\theta$

$$
\begin{aligned}
= & \left(\frac{3}{4\pi}\right)^{1/2}\left(\frac{5}{16\pi}\right)^{1/2}(2\pi)\int_1^{-1} t^2(3t^2-1) \\
= & \left(\frac{3}{4\pi}\right)^{1/2}\left(\frac{5}{16\pi}\right)^{1/2}(2\pi)\int_{-1}^{1}(3t^4-t^2)(dt)
\end{aligned}
$$

$3t^4-t^2$ 은 우함수이므로

$$
\begin{aligned}
= & \left(\frac{3}{4\pi}\right)^{1/2}\left(\frac{5}{16\pi}\right)^{1/2}(2\pi)(2)\int_0^{1}(3t^4-t^2)(dt) \\
= & \left(\frac{3}{4\pi}\right)^{1/2}\left(\frac{5}{16\pi}\right)^{1/2}(2\pi)(2)\left[\frac{3}{5}t^5-\frac{1}{3}t^3\right]_0^1 \\
= & \left(\frac{3}{4\pi}\frac{5}{16\pi}4\pi^2 4\right)^{1/2}\frac{4}{15} \\
= & \left(\frac{15}{4}\right)^{1/2}\frac{4}{15} \\
= & \left(\frac{4}{15}\right)^{1/2}
\end{aligned}
$$